

Georgios Karamousalis

JAVA & BACKEND ENGINEERING

Patras, Greece | +30 694 433 0566 | dgiorgoskar@gmail.com

[GitHub](#) | [Portfolio](#) | [LinkedIn](#)

PROFESSIONAL SUMMARY

Computer Science graduate with an MSc dissertation submitted in Artificial Intelligence. Builds Java and Spring Boot REST APIs with SQL persistence, JWT authentication and automated tests. Developed OrderOps with transactional inventory workflows and a Python forecasting service. Seeking a Junior Java / Backend Engineer role.

TECHNICAL SKILLS

Languages: Java, Python, SQL, JavaScript, TypeScript

Backend and data: Spring Boot, Spring Security, REST APIs, JWT, JPA/Hibernate, PostgreSQL, MySQL, MariaDB, FastAPI, Node.js/Express

Testing and delivery: JUnit, Testcontainers, Pytest, Playwright, Git, GitHub Actions, Docker, Maven

AI and frontend: PyTorch, Hugging Face Transformers, scikit-learn, pandas, NetworkX, React

SOFTWARE ENGINEERING PROJECTS

OrderOps

[GitHub](#)

Java 21, Spring Boot, PostgreSQL, JPA, JWT, React, Python, Docker

- Implemented multi-warehouse stock reservations, order fulfillment/cancellation and supplier receipts with database transactions, audit records and a durable notification outbox.
- Passed 29 automated tests across backend, forecasting and browser workflows, including 8 buyers competing for 1 stock unit; enforced server-side access for 3 roles.
- Integrated a forecasting API evaluated on 374 daily UCI retail observations: 29.45-unit MAE versus 32.86 for a seasonal baseline on a 14-day validation window.

LedgerGuard — Secure Wallet Platform

[GitHub](#)

Java 21, Spring Boot, JPA, PostgreSQL, React, Docker

- Built JWT-secured wallet APIs with transactional transfers, idempotency keys, optimistic locking and Flyway migrations; included JUnit, MockMvc and PostgreSQL Testcontainers tests.

ResolveHub — Support Ticket and SLA Platform

[GitHub](#)

Java 21, Spring Boot, PostgreSQL, React, Kafka

- Implemented ticket assignment, workflow transitions, priority-based SLA deadlines and an activity timeline with 3 access roles, lifecycle events and automated tests.

SupplyPilot — Retail Operations Platform

[GitHub](#)

Java 21, Spring Boot, PostgreSQL, React, Redis, RabbitMQ

- Developed inventory, supplier, expense and purchase-order workflows behind a JWT-secured API, with database migrations, Docker Compose and a monitoring configuration.

ADDITIONAL SOFTWARE PROJECTS

CareerFlow — Job Application Tracker

[GitHub \(private\)](#)

React, TypeScript, FastAPI, PostgreSQL — Built an application-tracking dashboard and a separate JWT-secured API with lexical resume-fit scoring; UI/API integration remains future work.

Restaurant Reservation App

[GitHub](#)

React, Node.js, Express, MariaDB — Built JWT login and reservation management across 3 relational tables; supplied database setup, sample data and a Postman collection.

Secure Task API

[GitHub \(private\)](#)

Java, Spring Boot, MySQL — Implemented validated task CRUD endpoints, JWT authentication, Swagger documentation, JUnit integration tests and Docker delivery.

Secure Notes API

[GitHub \(private\)](#)

Python, FastAPI, PostgreSQL — Built async, owner-scoped notes endpoints with JWT, Argon2 password hashing, typed validation, Pytest and Docker.

Web Quality Pipeline

[GitHub \(private\)](#)

Python, Selenium, Pytest — Created smoke, basic accessibility and same-origin link checks with HTML/JUnit reports, Docker execution and GitHub Actions.

gDates — Appointment Web Application

[GitHub \(private\)](#)

Python, Flask, SQLite — Built appointment submission, appointment-list and administration pages with session-based access controls.

Personal Portfolio Website

[GitHub](#)

React, Vite, CSS — Built a responsive portfolio presenting engineering projects and research, with downloadable CV, automated checks and GitHub Pages deployment.

RESEARCH PROJECTS

MSc Dissertation — Stance Detection with Deep Learning

- Compared TF-IDF/Logistic Regression, BERT and RoBERTa across VAST and EZ-STANCE using PyTorch and Transformers. Best reported EZ-STANCE result: 80.3% accuracy and Macro-F1; included error analysis and Streamlit.

BSc Dissertation — Social Network Analysis for Influencer Detection

- Used Python, NetworkX and Gephi to apply PageRank, centrality and community detection for influencer identification.

EDUCATION

MSc Artificial Intelligence | Dissertation submitted; graduation pending

BSc Computer Science | Metropolitan College, Athens | 2022–2025

ADDITIONAL EXPERIENCE

Hellenic Army — Expenditure and Supply Office: Maintained expense records and supply documentation with accuracy and confidentiality.

Family Wine Shop — Retail Operations: Supported customers, stock organization and daily operations.

LANGUAGES

Greek: Native | English: C2 / Proficient | German: Basic